WEB TECHNOLOGIES — MIDTERM COMPLETION DOCUMENT

The Brown Vanity | Tasks 1, 2 and 3

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Course: CSC336 — Web Technologies

Project: The Brown Vanity

Individual parameters

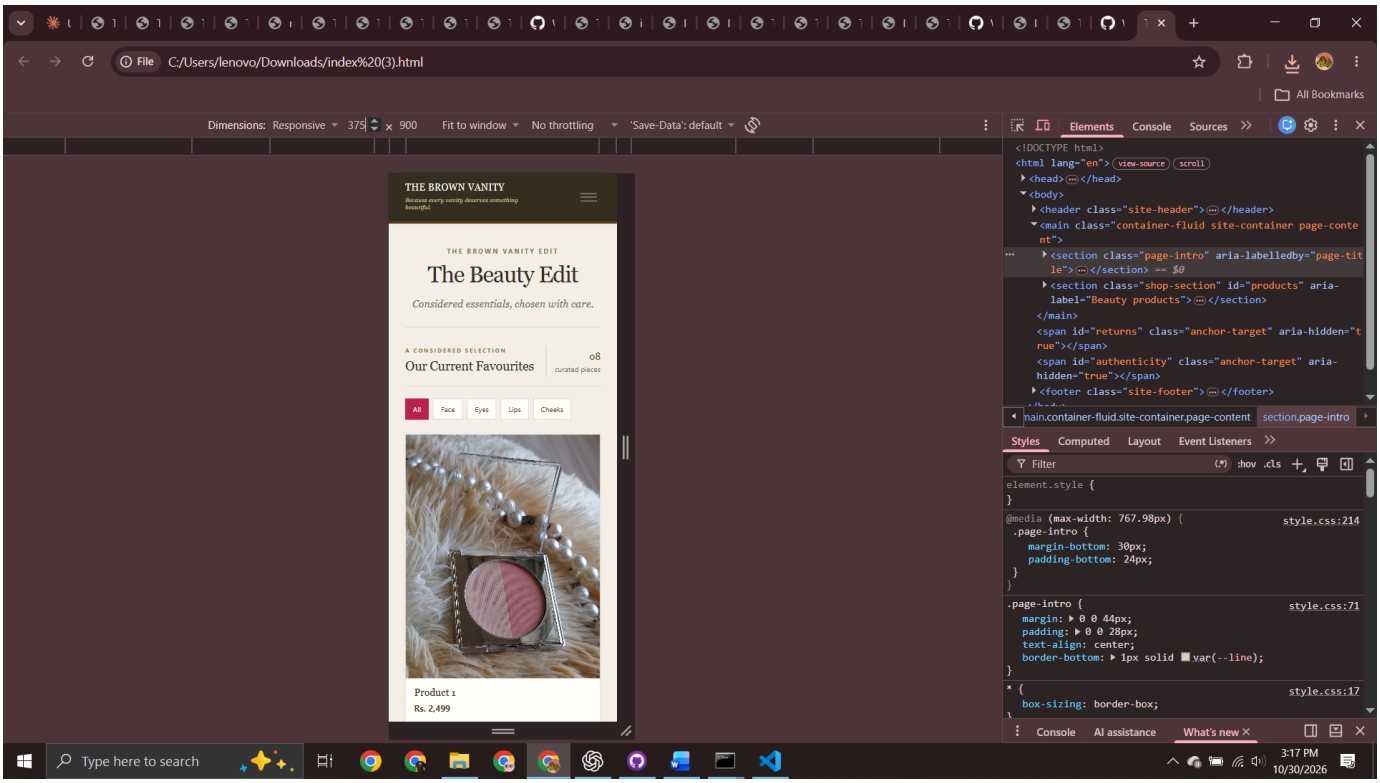
N = 49 | BP = 649px | H = 343 | K = 4

Sticky details offset = 79px | Premium threshold = Rs. 4,900

The following browser screenshots document the implementation at 375px, the required breakpoint (649px), and 1280px. The source code added or changed for each task is included as selectable text after the screenshots.

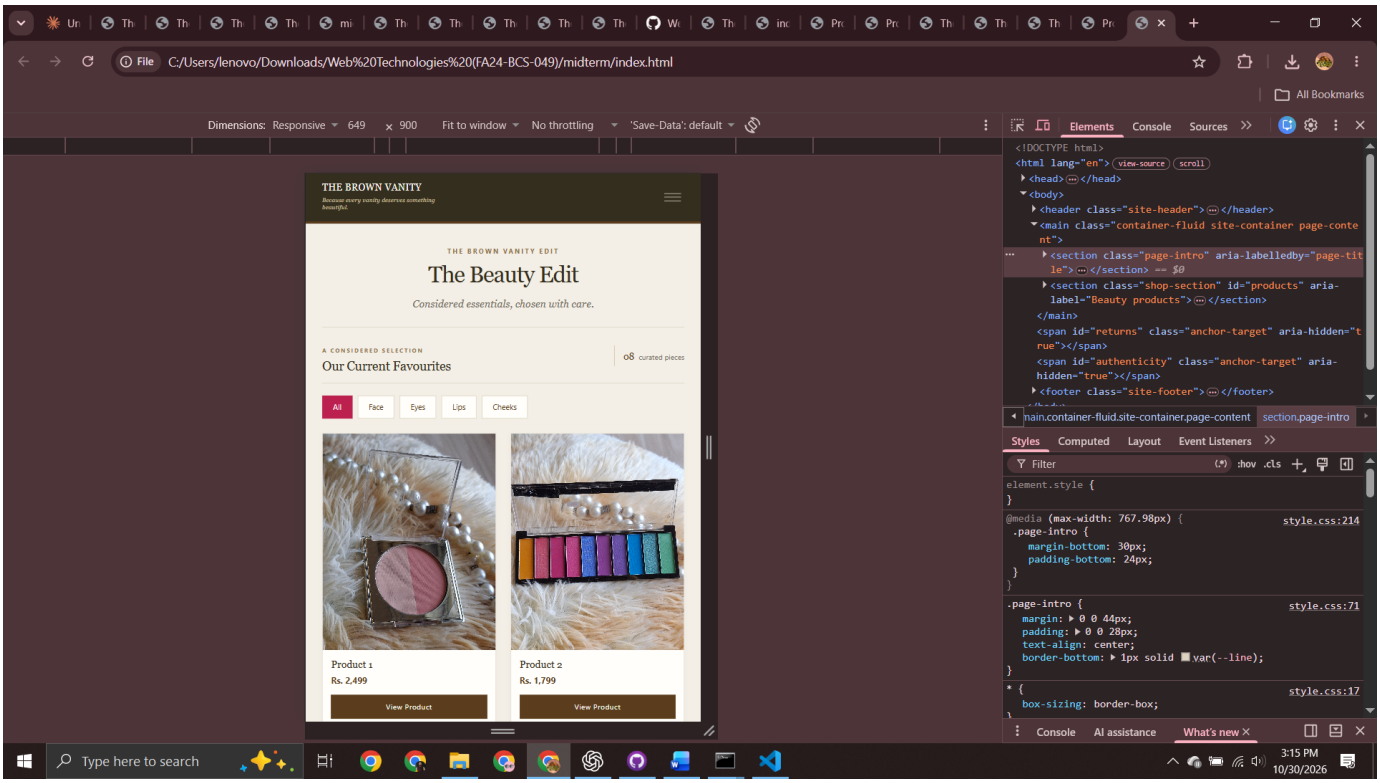
Task 1 — CSS-only category filter

375px — mobile layout



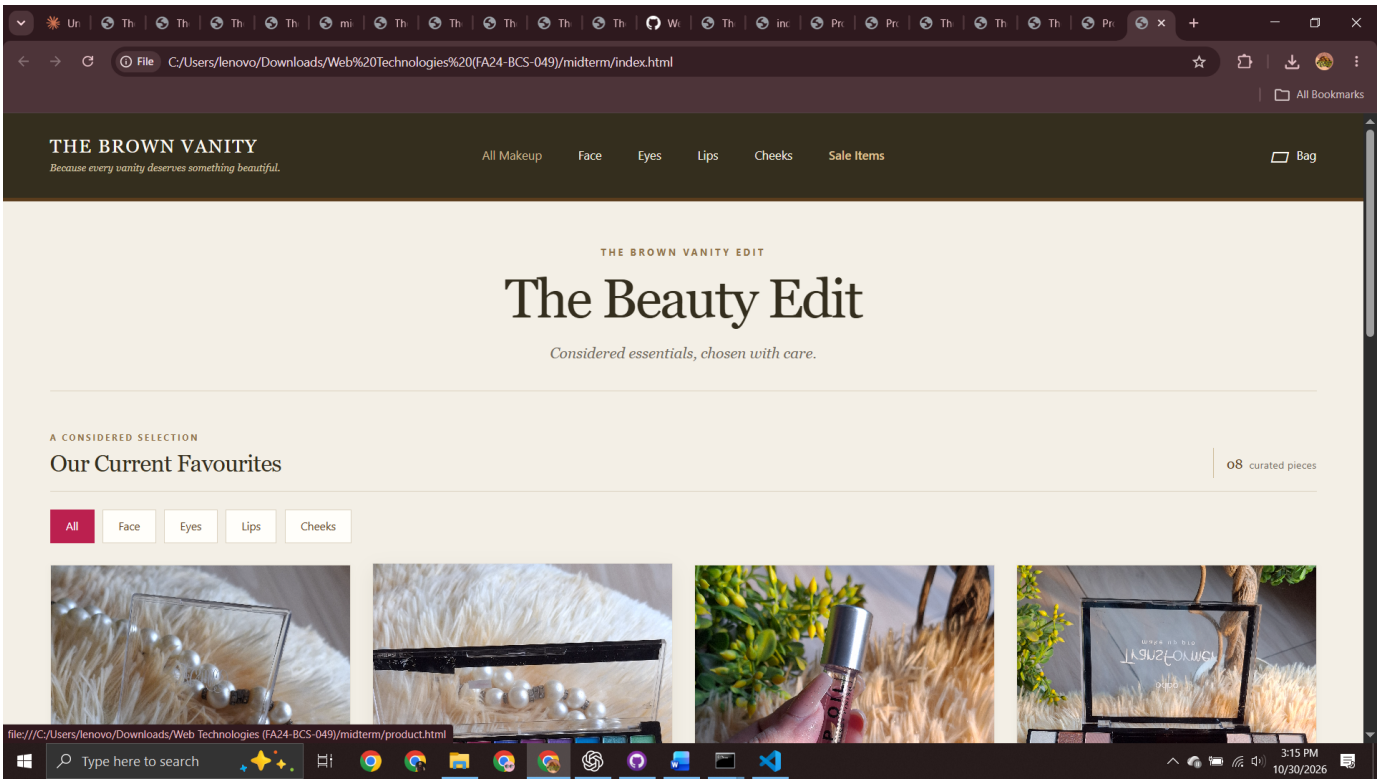
Task 1 — CSS-only category filter

649px — required breakpoint



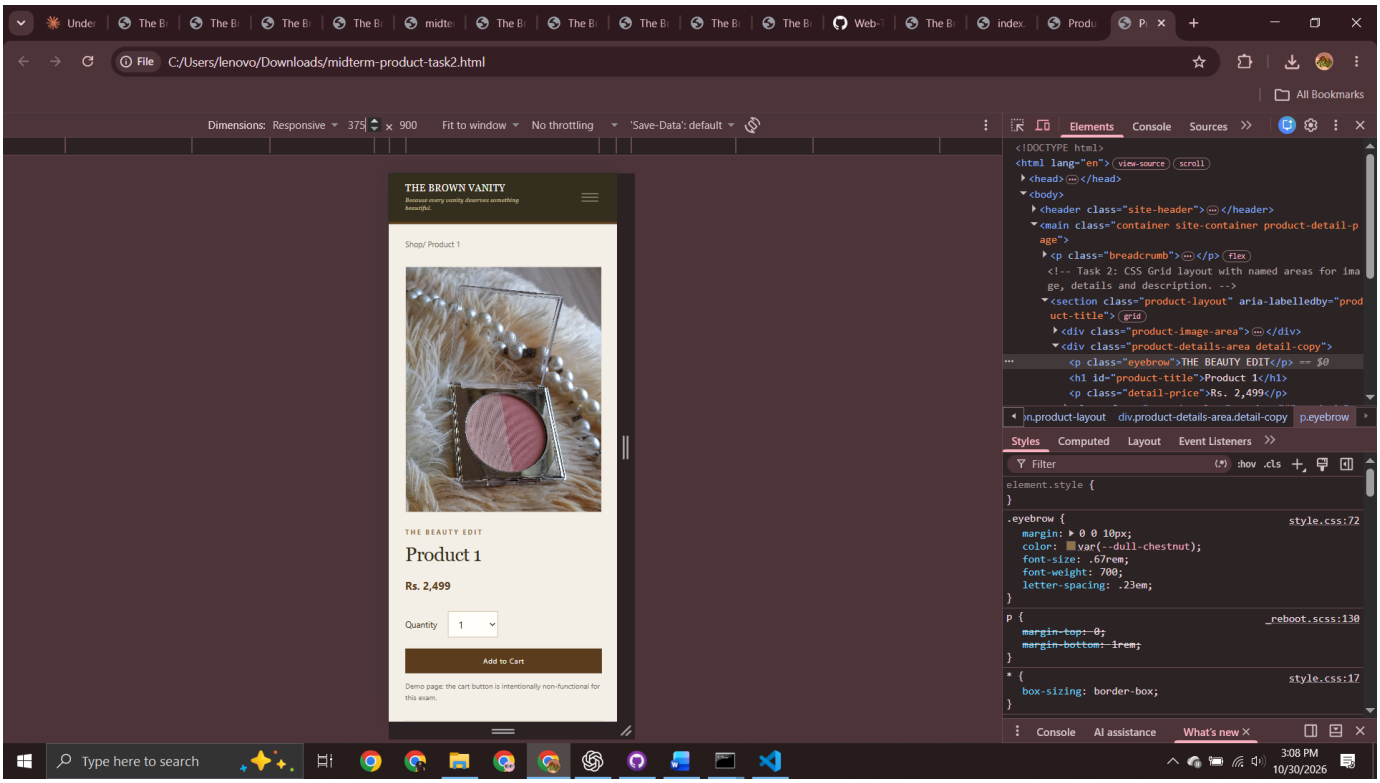
Task 1 — CSS-only category filter

1280px — desktop layout



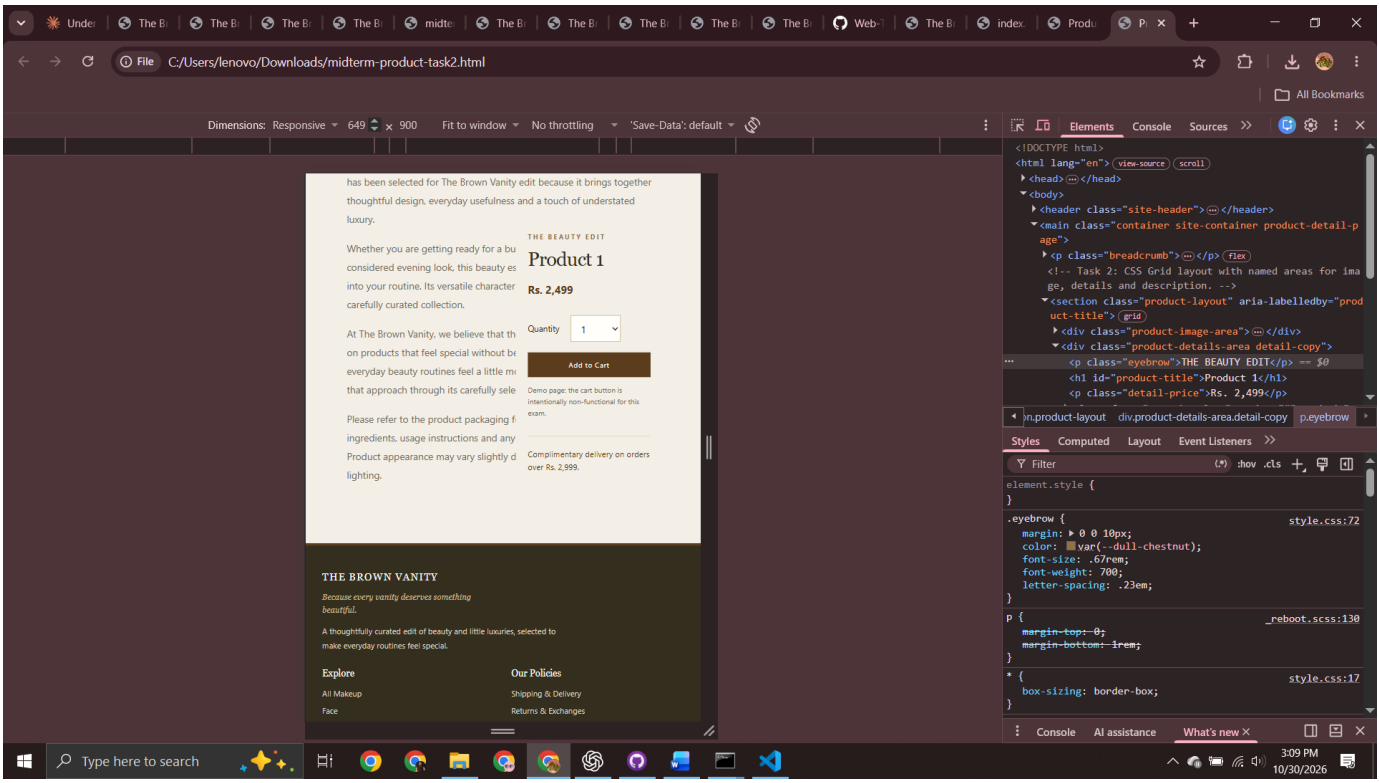
Task 2 — Product detail CSS Grid and sticky details

375px — mobile layout



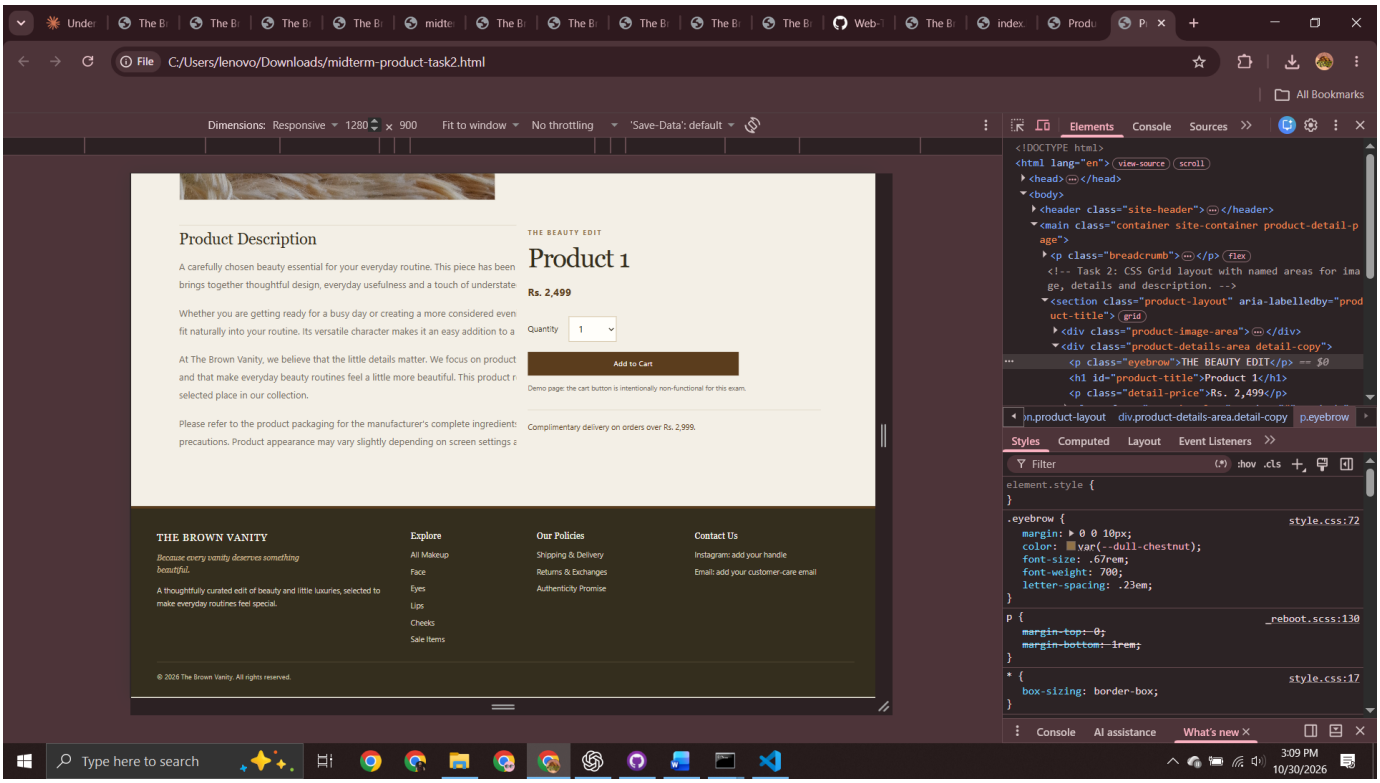
Task 2 — Product detail CSS Grid and sticky details

649px — required breakpoint



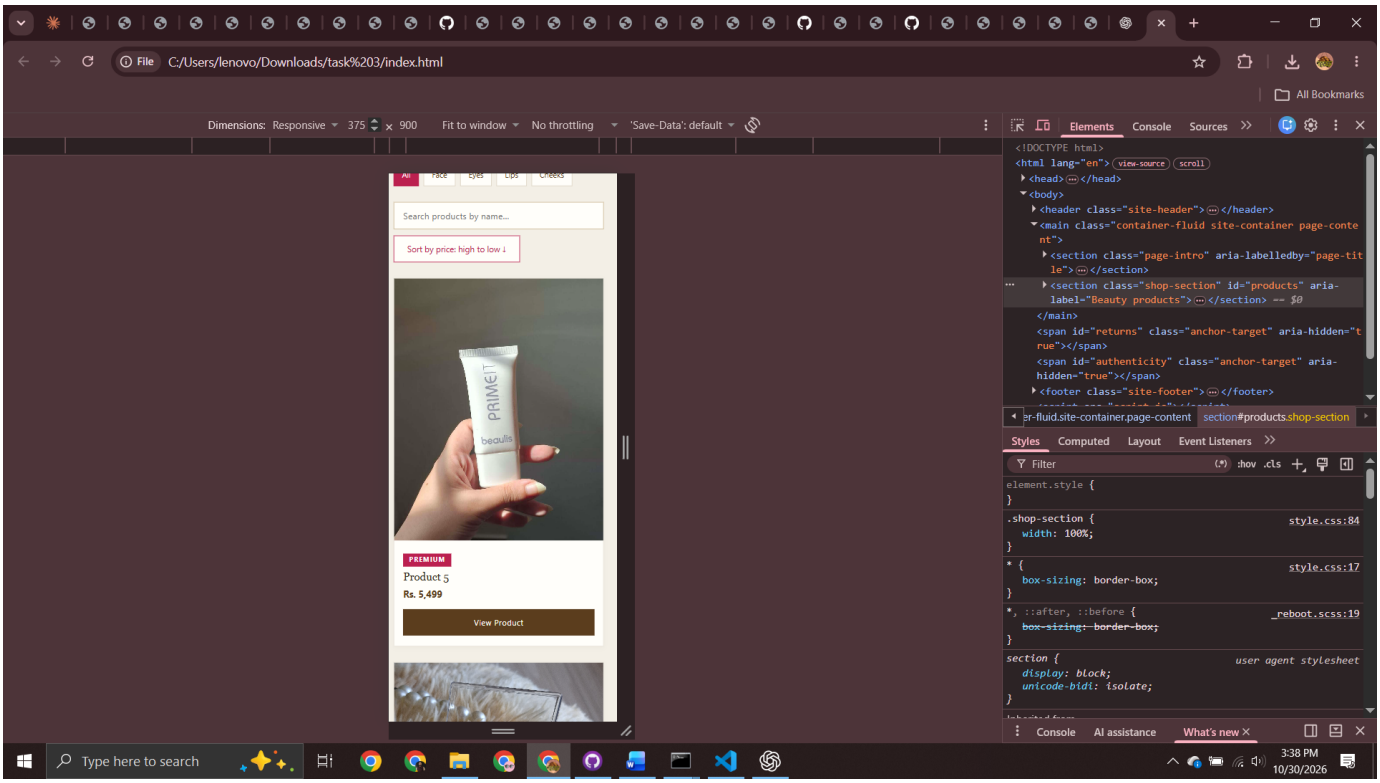
Task 2 — Product detail CSS Grid and sticky details

1280px — desktop layout



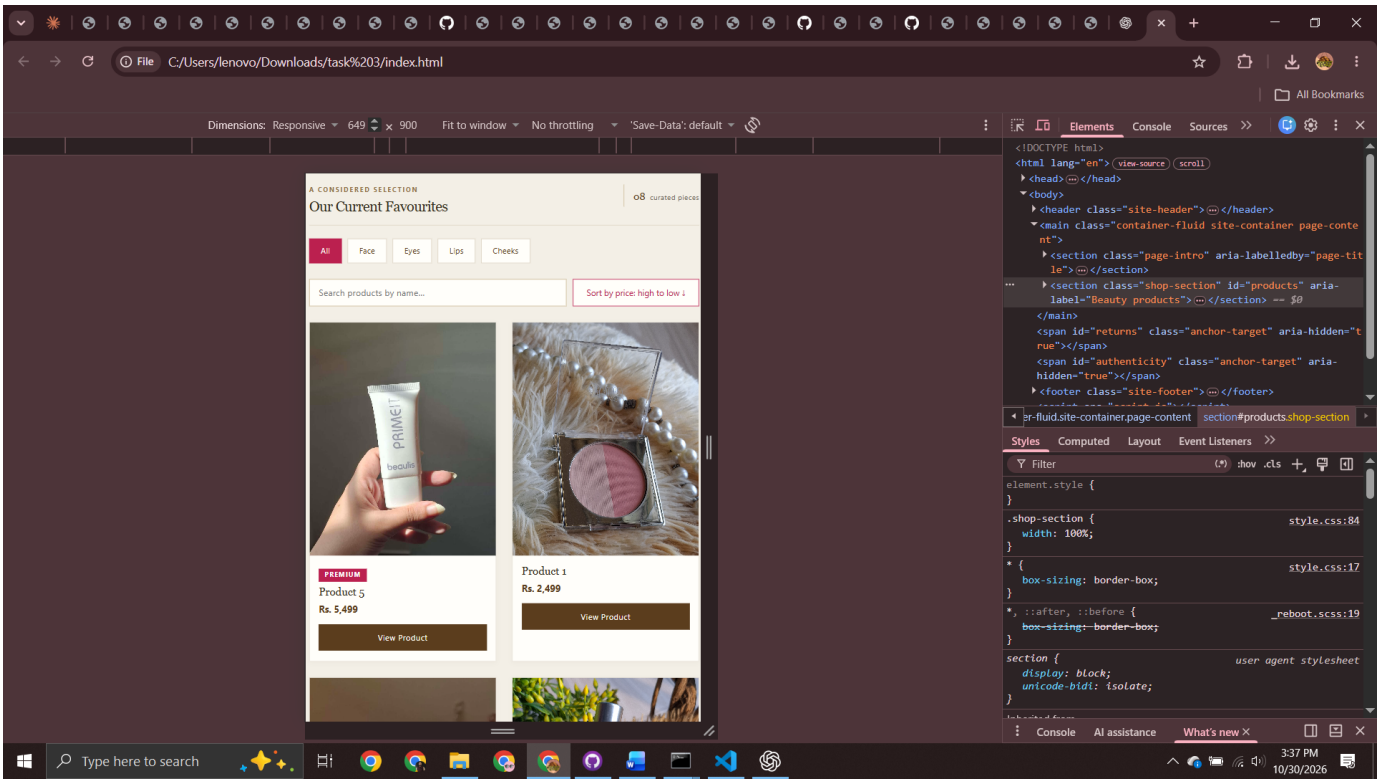
Task 3 — Live search, price sorting and Premium label

375px — mobile layout; Premium product visible



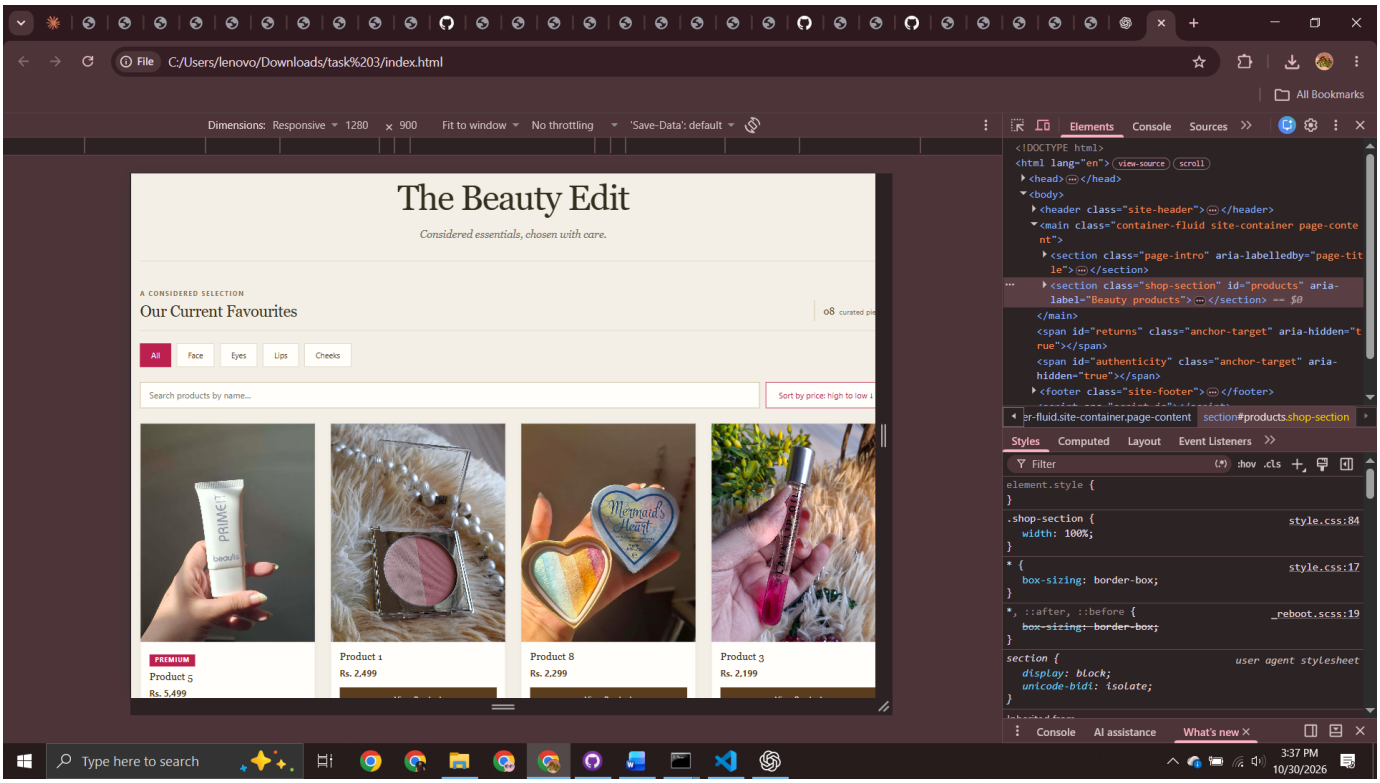
Task 3 — Live search, price sorting and Premium label

649px — required breakpoint; Premium product visible



Task 3 — Live search, price sorting and Premium label

1280px — sorted product list and Premium label visible



The 1280px screenshot shows the product cards sorted from high to low, with Product 5 priced at Rs. 5,499 and marked Premium (threshold: Rs. 4,900).

CODE ADDED OR CHANGED

The following excerpts are copied as text from the submitted HTML, CSS and JavaScript files.

Task 1 — CSS-only category filter

index.html — category radio controls and category classes on the eight existing product cards

```
      <!-- Task 1: CSS-only category filter. The radio inputs control the cards without JavaScript. -->
      <div class="category-filter" role="group" aria-label="Filter products by category">
        <input class="filter-radio" type="radio" name="category" id="filter-all" checked>
        <label class="filter-label" for="filter-all">All</label>

        <input class="filter-radio" type="radio" name="category" id="filter-face">
        <label class="filter-label" for="filter-face">Face</label>

        <input class="filter-radio" type="radio" name="category" id="filter-eyes">
        <label class="filter-label" for="filter-eyes">Eyes</label>

        <input class="filter-radio" type="radio" name="category" id="filter-lips">
        <label class="filter-label" for="filter-lips">Lips</label>

        <input class="filter-radio" type="radio" name="category" id="filter-cheeks">
        <label class="filter-label" for="filter-cheeks">Cheeks</label>
      </div>

<!-- Category classes applied to the existing eight product columns -->
<div class="col-12 col-sm-6 col-lg-3 product-col category-face">   <!-- Product 1 -->
<div class="col-12 col-sm-6 col-lg-3 product-col category-eyes">   <!-- Product 2 -->
<div class="col-12 col-sm-6 col-lg-3 product-col category-lips">   <!-- Product 3 -->
<div class="col-12 col-sm-6 col-lg-3 product-col category-cheeks"> <!-- Product 4 -->
<div class="col-12 col-sm-6 col-lg-3 product-col category-face">   <!-- Product 5 -->
<div class="col-12 col-sm-6 col-lg-3 product-col category-eyes">   <!-- Product 6 -->
<div class="col-12 col-sm-6 col-lg-3 product-col category-lips">   <!-- Product 7 -->
<div class="col-12 col-sm-6 col-lg-3 product-col category-cheeks"> <!-- Product 8 -->
```

style.css — category filter styling and CSS-only visibility rules

```
/* TASK 1 — CSS-only category filter */
.category-filter {
  display: flex;
  flex-wrap: wrap;
  align-items: center;
  gap: 9px;
  margin: 0 0 24px;
}

/* Keep radio inputs available to keyboard users and assistive technology. */
.filter-radio {
  position: absolute;
  width: 1px;
  height: 1px;
  padding: 0;
  margin: -1px;
  overflow: hidden;
  clip: rect(0, 0, 0, 0);
  white-space: nowrap;
  border: 0;
}

.filter-label {
  display: inline-block;
  margin: 0;
  padding: 8px 17px;
  border: 1px solid var(--line);
  background: var(--card);
  color: var(--soft-leather);
  font-size: .78rem;
  cursor: pointer;
  transition: background .18s ease, color .18s ease, border-color .18s ease;
}

/* Style the label immediately following the selected radio button. */
.filter-radio:checked + .filter-label {
  border-color: hsl(343, 70%, 45%);
  background: hsl(343, 70%, 45%);
  color: #FFFFFF;
}

.filter-radio:focus-visible + .filter-label {
  outline: 2px solid var(--pine-bark);
  outline-offset: 3px;
}

/* :has() detects the selected category and hides other complete product columns. */
.shop-section:has(#filter-face:checked) .product-grid .product-col:not(.category-face),
.shop-section:has(#filter-eyes:checked) .product-grid .product-col:not(.category-eyes),
```

```
.shop-section:has(#filter-lips:checked) .product-grid .product-col:not(.category-lips),
.shop-section:has(#filter-cheeks:checked) .product-grid .product-col:not(.category-cheeks) {
  display: none;
}

@media (max-width: 420px) {
  .category-filter {
    gap: 7px;
  }

  .filter-label {
    padding: 7px 12px;
    font-size: .74rem;
  }
}
```

Task 2 — Product detail page layout

product.html — named grid-area structure for image, details and description

```
    <!-- Task 2: CSS Grid layout with named areas for image, details and description. -->
    <section class="product-layout" aria-labelledby="product-title">

      <div class="product-image-area">
        <div class="detail-image-wrap">
          <img src="images/picture1.jpg" alt="Product 1" class="detail-image">
        </div>
      </div>

      <div class="product-details-area detail-copy">
        <p class="eyebrow">THE BEAUTY EDIT</p>
        <h1 id="product-title">Product 1</h1>
        <p class="detail-price">Rs. 2,499</p>

        <form class="quantity-form" action="#" method="get">
          <label for="quantity">Quantity</label>
          <select id="quantity" name="quantity">
            <option value="1">1</option>
            <option value="2">2</option>
            <option value="3">3</option>
          </select>
        </form>

        <button class="btn btn-brown btn-add-detail" type="button">Add to Cart</button>
        <p class="static-note">Demo page: the cart button is intentionally non-functional for this exam.</p>
        <p class="delivery-inline">Complimentary delivery on orders over Rs. 2,999.</p>
      </div>

      <div class="product-description-area">
        <h2>Product Description</h2>
        <p class="detail-description">
          A carefully chosen beauty essential for your everyday routine. This piece has been
          selected for The Brown Vanity edit because it brings together thoughtful design,
          everyday usefulness and a touch of understated luxury.
        </p>
        <p class="detail-description">
          Whether you are getting ready for a busy day or creating a more considered evening
          look, this beauty essential is designed to fit naturally into your routine. Its
          versatile character makes it an easy addition to a carefully curated collection.
        </p>
        <p class="detail-description">
          At The Brown Vanity, we believe that the little details matter. We focus on products
          that feel special without being excessive, and that make everyday beauty routines
          feel a little more beautiful. This product reflects that approach through its
          carefully selected place in our collection.
        </p>
        <p class="detail-description">
          Please refer to the product packaging for the manufacturer's complete ingredients,
          usage instructions and any product-specific precautions. Product appearance may
          vary slightly depending on screen settings and lighting.
        </p>
      </div>
```

style.css — CSS Grid areas, 649px breakpoint and sticky details box

```
/* TASK 2 — Product page CSS Grid layout with named grid areas. */
.product-layout {
  display: grid;
  grid-template-areas:
    "image"
    "details"
    "description";
  grid-template-columns: minmax(0, 1fr);
  gap: 28px;
  align-items: start;
}

.product-image-area {
  grid-area: image;
  min-width: 0;
}

.product-details-area {
  grid-area: details;
  min-width: 0;
}

.product-description-area {
  grid-area: description;
  min-width: 0;
  padding-top: 8px;
  border-top: 1px solid var(--line);
}

.product-description-area h2 {
  margin: 0 0 16px;
```

```
  color: var(--pine-bark);
  font-family: Georgia, "Times New Roman", serif;
  font-size: 1.65rem;
  font-weight: 400;
}

.product-description-area .detail-description {
  max-width: 900px;
  margin: 0 0 18px;
  color: var(--muted);
  line-height: 1.9;
}

/* At 649px and wider, place image and details side by side.
   The description spans both columns on the next row. */
@media (min-width: 649px) {
  .product-layout {
    grid-template-areas:
      "image details"
      "description description";
    grid-template-columns: minmax(0, 1fr) minmax(0, 1fr);
    column-gap: 36px;
    row-gap: 42px;
  }

  /* Keep the details visible while the user scrolls through the page. */
  .product-details-area {
    position: sticky;
    top: 79px;
    align-self: start;
    background: var(--paper);
    z-index: 5;
    padding: 20px;
}
}

/* Below 649px, the named areas stack in the required order.
   Sticky positioning is disabled on the narrow layout. */
@media (max-width: 648.98px) {
  .product-layout {
    grid-template-areas:
      "image"
      "details"
      "description";
    grid-template-columns: minmax(0, 1fr);
    gap: 24px;
  }

  .product-details-area {
    position: static;
  }
}

/* Task 2 (extra): a taller image gives the sticky details box room to stay visible while scrolling. */
.product-image-area .detail-image { max-height: none; aspect-ratio: 3 / 4; }
```

Task 3 — JavaScript search, sorting and Premium label

index.html — search controls, no-results message, Product 5 price and script link

```
      <!-- Task 3: live search and price sort (script.js). -->
      <div class="tools-bar">
        <input type="search" id="search-box" class="search-box" placeholder="Search products by name…"
          aria-label="Search products by name">
        <button type="button" id="sort-btn" class="btn-sort">Sort by price</button>
      </div>
      <p id="no-results" class="no-results" hidden>No products found</p>

<!-- Product 5 price used to demonstrate the Premium threshold -->
          <div class="col-12 col-sm-6 col-lg-3 product-col category-face">
            <article class="product-card">
              <a class="product-image-link" href="product.html" aria-label="View Product 5">
                <img src="images/picture5.jpg" alt="Product 5" class="product-image">
              </a>
              <div class="product-info">
                <h2><a href="product.html">Product 5</a></h2>
                <p class="product-price">Rs. 5,499</p>
                <a class="btn btn-brown" href="product.html">View Product</a>
              </div>
            </article>
          </div>

<!-- Load the separate plain JavaScript file -->
    </div>
  </footer>
  <script src="script.js"></script>
</body>
</html>
```

style.css — search, sort, no-results and Premium badge styles

```
/* TASK 3 — live search, price sort and Premium label */
[hidden] { display: none !important; }
.tools-bar { display: flex; flex-wrap: wrap; gap: 10px; margin: 0 0 24px; }
.search-box {
  flex: 1 1 220px; padding: 9px 14px; border: 1px solid var(--windy-dunes); border-radius: 0;
  background: var(--card); color: var(--pine-bark); font-size: .85rem;
}
.search-box:focus { outline: 2px solid hsl(343, 70%, 45%); outline-offset: 2px; }
.btn-sort {
  padding: 9px 20px; border: 1px solid hsl(343, 70%, 45%); border-radius: 0;
  background: var(--card); color: hsl(343, 70%, 45%); font-size: .8rem; cursor: pointer;
  transition: background .18s ease, color .18s ease;
}
.btn-sort:hover { background: hsl(343, 70%, 45%); color: #fff; }
.no-results { padding: 48px 0; text-align: center; color: var(--muted); font-family: Georgia, "Times New
  Roman", serif; font-size: 1.1rem; font-style: italic; }
.product-col.search-hidden { display: none; }   /* hidden by the search box (script.js) */
.premium-badge {
  display: inline-block; margin-bottom: 6px; padding: 2px 9px; background: hsl(343, 70%, 45%);
  color: #fff; font-size: .62rem; font-weight: 700; letter-spacing: .12em; text-transform: uppercase;
}
```

Task 3 — JavaScript source

script.js — complete plain JavaScript implementation

```
// Midterm Task 3 - plain JavaScript, works on the product cards already in the page.
const N = 49;
const PREMIUM_LIMIT = N * 100;                // 4900

const grid = document.querySelector('.product-grid');
const cards = Array.from(grid.querySelectorAll('.product-col'));
const searchBox = document.getElementById('search-box');
const sortBtn = document.getElementById('sort-btn');
const noResults = document.getElementById('no-results');
let ascending = true;                         // next sort direction

// Read the name and price straight from the card.
function getName(card) {
  return card.querySelector('.product-info h2').textContent.trim().toLowerCase();
}
function getPrice(card) {
  return parseInt(card.querySelector('.product-price').textContent.replace(/[^\d]/g, ''), 10);
}

// The checked radio id is "filter-all", "filter-face", ... -> "all", "face", ...
function getCategory() {
  const checked = document.querySelector('input[name="category"]:checked');
  return checked.id.replace('filter-', '');
}

// A card is shown only if it matches BOTH the search text and the category.
// Category hiding is done by CSS (Task 1); here we add the search hiding and
// count the cards matching both, to decide whether to show "No products found".
function applyFilters() {
  const text = searchBox.value.trim().toLowerCase();
  const category = getCategory();
  let visibleCount = 0;

  cards.forEach(function (card) {
    const matchesSearch = getName(card).includes(text);
    const matchesCategory = category === 'all' || card.classList.contains('category-' + category);
    card.classList.toggle('search-hidden', !matchesSearch);
    if (matchesSearch && matchesCategory) visibleCount++;
  });

  grid.hidden = visibleCount === 0;           // message replaces the grid
  noResults.hidden = visibleCount !== 0;
}

searchBox.addEventListener('input', applyFilters);
document.querySelectorAll('input[name="category"]').forEach(function (radio) {
  radio.addEventListener('change', applyFilters);
});

// Sort by price: 1st click low -> high, 2nd click high -> low, and so on.
// The existing card elements are moved with appendChild (no innerHTML).
sortBtn.addEventListener('click', function () {
  const sorted = Array.from(grid.querySelectorAll('.product-col')).sort(function (a, b) {
    return ascending ? getPrice(a) - getPrice(b) : getPrice(b) - getPrice(a);
  });
  sorted.forEach(function (card) { grid.appendChild(card); });
  sortBtn.textContent = ascending ? 'Sort by price: low to high \u2191' : 'Sort by price: high to low
    \u2193';
  ascending = !ascending;
});

// "Premium" label for products costing more than N x 100, added when the page loads.
cards.forEach(function (card) {
  if (getPrice(card) > PREMIUM_LIMIT) {
    const badge = document.createElement('span');
    badge.className = 'premium-badge';
    badge.textContent = 'Premium';
    card.querySelector('.product-info').prepend(badge);
  }
});
```